



Factor GCN for End-to-End Portfolio Optimization

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Motivation

Asset allocation seeks an optimal capital weight vector $\mathbf{w}$ to maximize risk-adjusted performance, fundamentally formulated as the Sharpe Ratio :

$$\max_{\mathbf{w}} \text{SharpeRatio}(\mathbf{w}) = \frac{\text{Benefit}(\mathbf{w})}{\text{Risk}(\mathbf{w})} = \frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}} \\ \text{subject to } \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \succeq \mathbf{0}$$

- $\boldsymbol{\mu} \in \mathbb{R}^N$ (Expected Return) : The expected return vector, serving as the profit driver for the optimization model.
- $\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$ (Covariance Matrix) : The covariance matrix, quantifying asset volatility and the structural risk of the portfolio.
- $R_f \in \mathbb{R}$ (Risk-Free Rate) : The absolute baseline return without market risk, used to evaluate expected excess returns.

The ML Solution

- Traditional Flaw : Plug-in estimators ($\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}$) suffer from market noise and ill-conditioning.
- Factor GCN : An end-to-end non-linear solver directly mapping market graphs to the optimal simplex Δ^{N-1} for noise-resistant allocation.

Methodology STAGE 1

Factor-Based Structural Graph

We construct a market graph from long-term fundamental factors rather than historical return correlations.

$$A_{ij} = \exp \left(-\frac{\|z_i - z_j\|^2}{2\sigma^2} \right)$$

where z_i contains firm characteristics such as Market Capitalization, ROA, and Debt Ratio.

$$\tilde{A} = A + I_N, \quad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}, \quad \tilde{L}_{sym} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$$

The resulting normalized graph captures stable economic relationships between firms and serves as a structural filter for message passing, allowing short-term signals to propagate through fundamentally similar assets while suppressing noisy price correlations.

Methodology STAGE 2

Spectral Feature Fusion & Sparse Activation

We introduce short-term momentum signals (X) and leverage the structural map ($\tilde{L}_{sym}$) as a spatial low-pass filter. To extract high-level representations, we define Θ as our primary learnable spectral parameter.

$$H = \text{SparseAct}_K \left(\tilde{L}_{sym} X \Theta \right)$$

- $\Theta \in \mathbb{R}^{F \times F'}$: The learnable feature extraction matrix.
- $\text{SparseAct}_K(\cdot)$: A Top-K activation function that zeroes out all but the most promising K signals, maximizing the signal-to-noise ratio.

End-to-End Optimization via Gradient Descent

The filtered hidden state H is projected into an investment confidence vector $\mathbf{z}$ via a secondary scoring parameter Φ . The signals are mapped into a strict capital allocation vector $\mathbf{w}$ using a Temperature-scaled Softmax function.

- Scoring & Allocation: $\mathbf{z} = H\Phi$, $\mathbf{w} = \text{Softmax}(\mathbf{z}/\tau)$
- Objective Function: $\mathcal{L}(\Theta, \Phi) = -\text{SharpeRatio}(\mathbf{w}) = -\frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}}$

Backpropagation & The Training Loop:

Instead of traditional quadratic programming, we optimize Θ and Φ over **500** epochs using Gradient Descent. The gradients of the Sharpe Ratio flow backward through the network via the Chain Rule of Calculus:

$$\nabla_{\Phi} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathbf{w}} \cdot \frac{\partial \mathbf{w}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial \Phi} \quad \nabla_{\Theta} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathbf{w}} \cdot \frac{\partial \mathbf{w}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial H} \cdot \frac{\partial H}{\partial \Theta}$$

Through a learning rate η , both parameters are iteratively updated.

$$\Theta \leftarrow \Theta - \eta \nabla_{\Theta} \mathcal{L}, \quad \Phi \leftarrow \Phi - \eta \nabla_{\Phi} \mathcal{L}$$

Combined with an Early Stopping mechanism to prevent overfitting, this allows the model to autonomously adapt to true market dynamics.

Generalization and Future Work

- **Customizable Factor Inputs** : The framework is highly adaptable, allowing portfolio managers to flexibly substitute the underlying structural factors to align with their specific investment mandates and risk preferences.
- **Dynamic Perception via GAT** : Future iterations will integrate Graph Attention Networks (GAT) to dynamically learn and adjust edge weights, enabling real-time adaptation to sudden market regime shifts.

Empirical Results

The chart below illustrates the out-of-sample cumulative returns of Factor GCN versus traditional benchmarks:

- Universe & Data : Top 150 listed companies in Taiwan (TEJ Pro database).
- Period: 2024/01–2026/05 (Out-of-Sample).
- Baselines: Market (0050.TW), Equal-Weight, and Hierarchical Risk Parity.
- Execution: Monthly dynamic rebalancing without standard transaction costs incorporated.

